# PROPABILITY

Probability is a measure of how likely an event will happen.



1% chance of rain in AD



90% chance sunny in AD



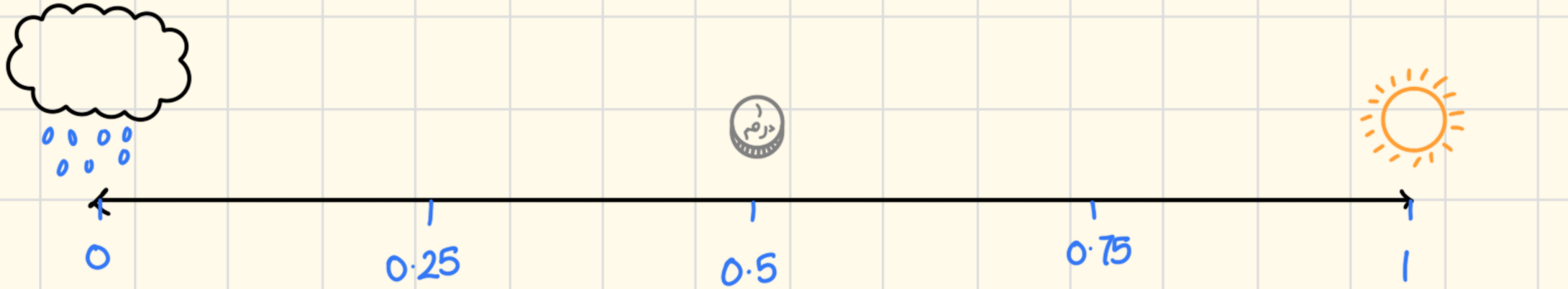
1/6 chance to get 2

# PROPABILITY

Probabilities lie between 0% and 100%.

OR

0 and 1.




Not going to happen

unlikely or not probable

equally likely

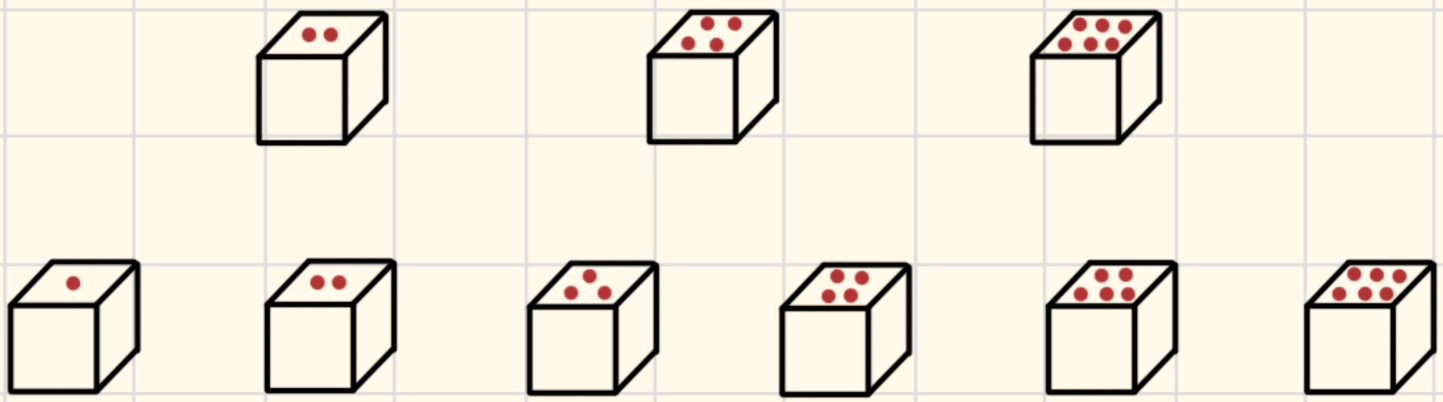
very likely probable

always gonna happen

# PROPABILITY

$$\text{Probability} = \frac{\#\text{ of ways to get event}}{\text{total \# of events}}$$

What's the probability of getting an even dice roll?



3 events

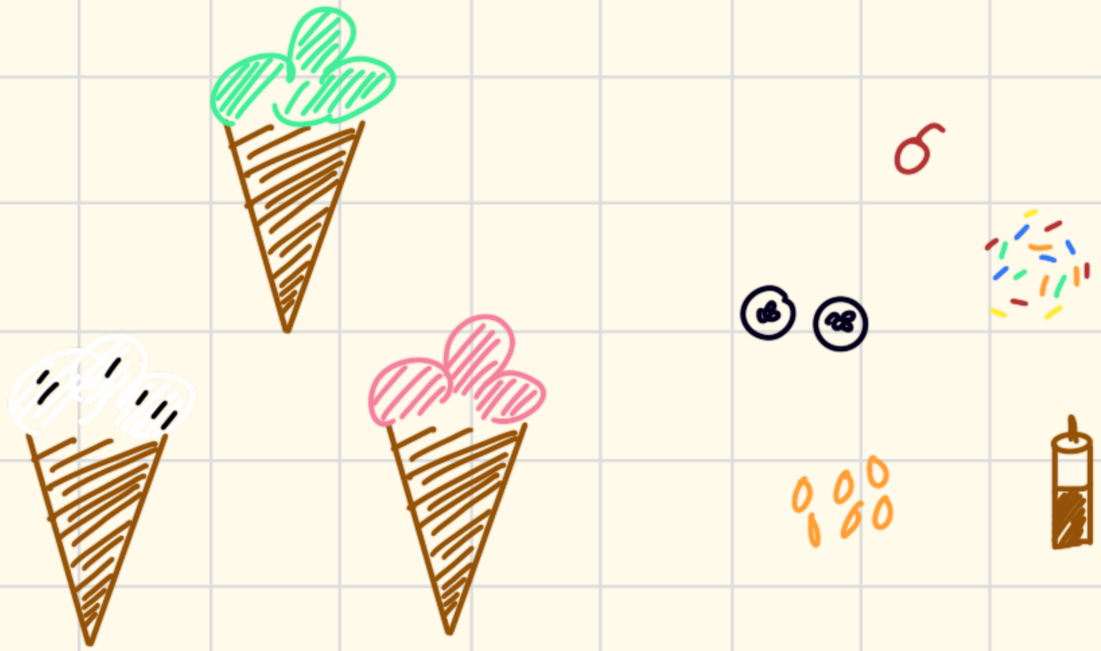
6 total

$$p = \frac{\#\text{ evens}}{\text{total}} = \frac{3}{6} = \frac{1}{2}$$

# Counting Events

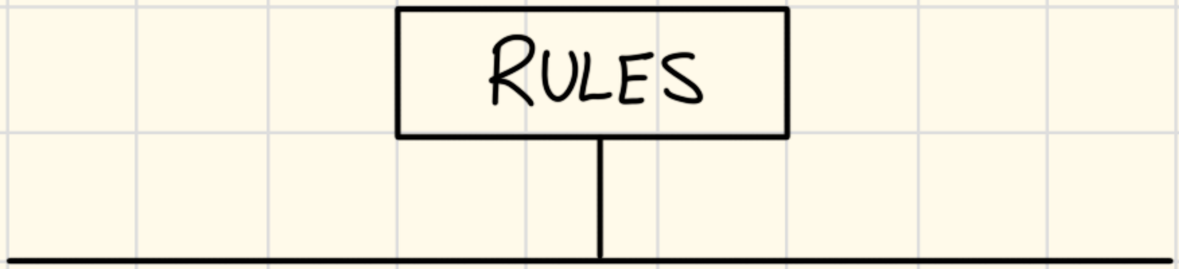
We can count total number of events by multiplying individual event outcomes.

How many different ice creams can you have if:



3 flavors × 5 toppings = 15 different ice cream combinations!

# Probability Road Map




## One Trial

### Mutually Exclusive

$P(A \text{ or } B) = P(A) + P(B)$

### Mutually Inclusive

$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

## Multiple Trials

### Independent

$P(A \text{ and } B) = P(A) \times P(B)$

### Dependent

$P(A \text{ and } B) = P(A) \times P(B \text{ given } A)$

# One Trial Events

Two events are mutually exclusive if they can't happen at the same time.



Pick red

OR green




Spin 1

OR 5



Toss dirham

OR dalla



Roll even

OR odd

# One Trial Events

## Mutually Exclusive

$$P(A \text{ or } B) = P(A) + P(B)$$

Find the probability of a



Pick red

OR green

$$\frac{3}{10} + \frac{2}{10}$$




Spin 1

OR 5

$$\frac{1}{6} + \frac{1}{6}$$

# One Trial Events

Two events are mutually inclusive if they have events in common.




Spin odd
OR greater than 4



Roll even
OR less than 4

# One Trial Events

## Mutually Inclusive

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

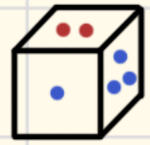



Spin odd

OR greater than 4

$$\frac{4}{8} + \frac{4}{8} - \frac{2}{8}$$

1, 3, 5, 7 — 5, 6, 7, 8 — 5, 7



Roll even

OR less than 3

$$\frac{3}{6} + \frac{2}{6} - \frac{1}{6}$$

2, 4, 6 — 1, 2 — 2

# Multiple Trial Events

Two events are independent if one event doesn't affect the probability of another.



Pick red

then pick green

* replacement



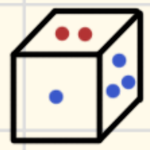

Spin 1

then spin 5



Toss dirham

then toss dalla



Roll even

then roll odd

# Multiple Trial Events

## Independent

$$P(A \text{ and } B) = P(A) \times P(B)$$



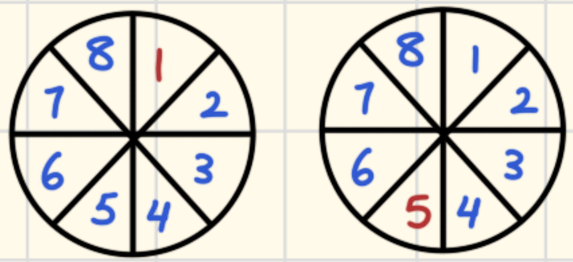
Pick red

then pick green

* replacement

$$\frac{3}{10} \times \frac{2}{10}$$

$$0 \qquad 0$$



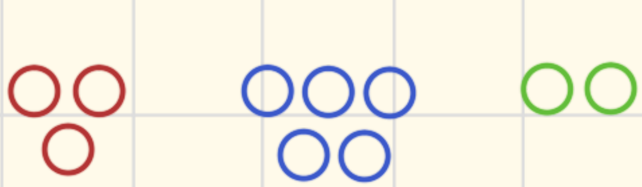

Spin 1

then spin 5

$$\frac{1}{8} \times \frac{1}{8}$$

# Multiple Trial Events

Two events are dependent if one event affects the probability of another.



Pick red
then pick green

without replacement



Pick girl
then pick boy

without replacement

# Multiple Trial Events

## Dependent

$$P(A \text{ and } B) = P(A) \times P(B \text{ given } A)$$



Pick red
then pick green
w/o replacement

$$\frac{3}{10} \times \frac{2}{9}$$

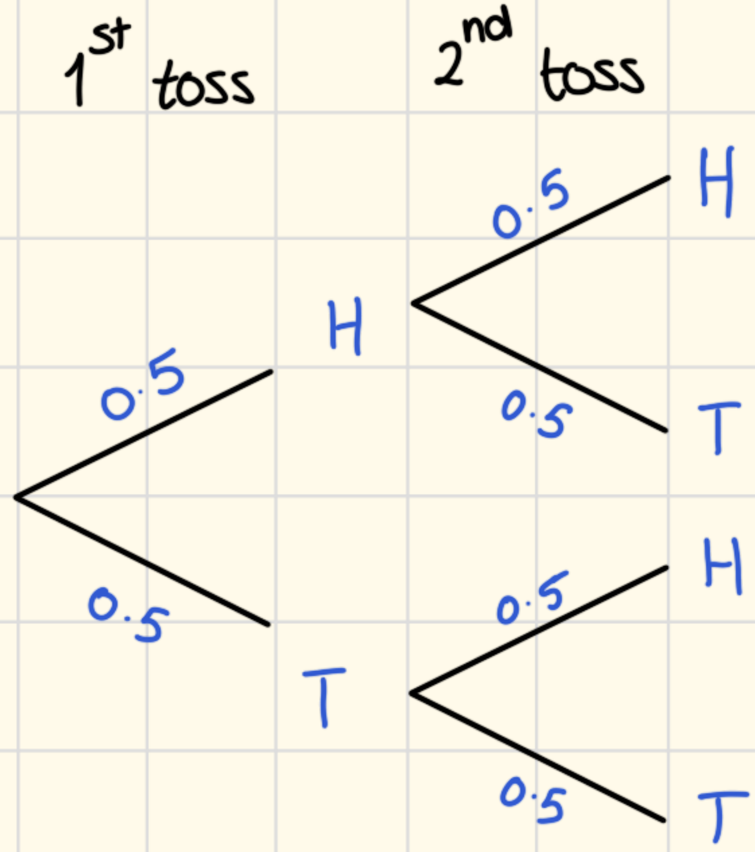


Pick girl
then pick boy
w/o replacement

$$\frac{3}{5} \times \frac{2}{4}$$

# TREE DIAGRAMS

Tree diagrams are awesome for a combination of events.



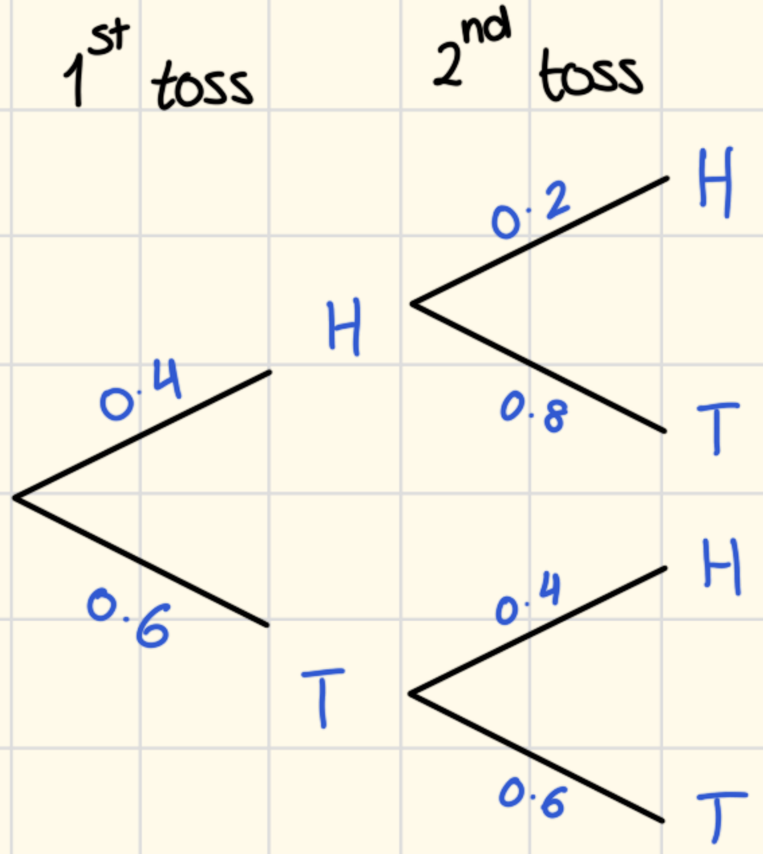

We can answer many questions!

What's the probability of

- head → head
- no heads
- tails in 2nd toss
- At least 1 tails

# TREE DIAGRAMS

We multiply in branches.
We add across branches.




Find probability of :

a) head → head : $P(H) \times P(H \text{ given } H)$

$0.4 \times 0.2$

b) no heads : $P(\text{no } H) \times P(\text{no } H \text{ given no } H)$

$0.6 \times 0.6$

c) tails in 2nd toss : $P(H) \times P(T \text{ given } H) + P(T) \times P(T \text{ given } T)$

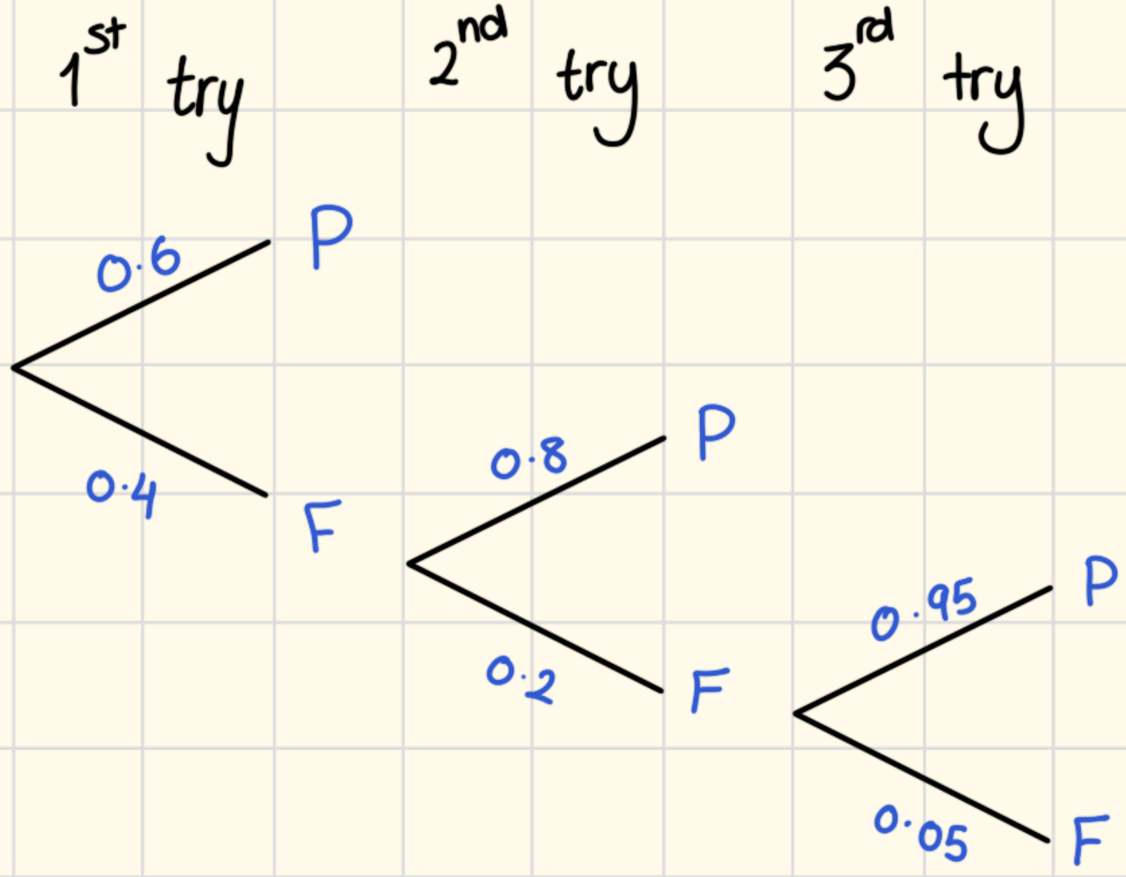
$0.4 \times 0.8 + 0.6 \times 0.6$

d) At least 1 tails : $1 - P(H) \times P(H \text{ given } H)$

$1 - 0.4 \times 0.2$

# TREE DIAGRAMS

Sally takes a driving test.




Find probability of :

a) She passes 1st try $0 \cdot 6$

2nd try $0 \cdot 4 \times 0 \cdot 8$

3rd try $0 \cdot 4 \times 0 \cdot 2 \times 0 \cdot 95$

b) She passes any try $0 \cdot 6$

$+ \quad 0 \cdot 4 \times 0 \cdot 8$

$+ \quad 0 \cdot 4 \times 0 \cdot 2 \times 0 \cdot 95$